

Week 11, November 20, 2024

Main topics

- Linearization of nonlinear inverse problems
- Nonlinear Tikhonov and Landweber

Material

From last week: Kirsch 4-4.1, Appendix A.7

New material: Kirsch 4.2 (pp. 122-123) and 4.3 (pp. 149-151, Theorem 4.37 (p. 154) and Theorem 4.44 (p. 161))

Note that we are only scratching the surface of a huge and deep topic regarding the mathematical analysis and computations for nonlinear inverse problems. This is a hot research topic that you might run into later.

Program

13:00-13:30 Nonlinear inverse problems and linearization

13:30-15:45 Exercises (see below)

16:00-17:00 Lecture: Optimization and iterative methods for nonlinear inverse problems

Exercises

1. Please visit <https://evaluating.dtu.dk/> and share your thoughts on the course. In particular: what can be improved and what should remain.
2. Consider the autoconvolution operator K in $L^2(0, 1)$ defined by

$$K(x)(t) = \int_0^t x(t-s)x(s) \, ds, \quad s \in (0, 1).$$

The linearization $K'[x_0] \in B(L^2(0, 1))$ of K around $x_0 \in L^2(0, 1)$ is

$$K'[x_0]h(t) = 2 \int_0^t x_0(t-s)h(s) \, ds, \quad t \in (0, 1).$$

In this exercise we will numerically approximate a solution to the nonlinear problem by means of linearization.

- (a) Last week you implemented numerically the autoconvolution mapping. (In case you were not succesful, I have uploaded pieces of code.)

Fix the function $x(s) = \exp(-s)$ and compute $y(t) = K(x)(t)$. Moreover, discretize $x(s)$ numerically and compute using the code a discrete approximation of $y(t)$.

- (b) Fix the function $x_0(s) = (e^{-1} - 1)s + 1$. Set up a matrix approximation A of $K'[x_0]$ and approximate using Tikhonov regularization a solution to the linearized problem

$$Ah = b + n,$$

where b is a discrete approximation of $y - K(x_0)$ and n is a random vector i.i.d. with relative noise level 5%.

- (c) Experiment with the above for other functions and other noise levels.

3. In Electrical Impedance Tomography, introduce $h = \gamma - 1$. Write a first order approximation

$$\Lambda_\gamma - \Lambda_1 \approx D\Lambda[1] h$$

where $D\Lambda[1]$ is linear (the Fréchet derivative of Λ at $\gamma = 1$). To make notation simpler we write $L = D\Lambda[1]$.

One can show that

$$((Lh)g_1, g_2)_{L^2(\partial\Omega)} = - \int_{\Omega} h \nabla v_1 \cdot \nabla \overline{v_2} \, dx \quad (1)$$

with v_j satisfying $\Delta v_j = 0$ in Ω and $\partial_\nu v_j|_{\partial\Omega} = g_j$ (ν denotes the outward unit normal). The complex conjugate is necessary since we will shortly use complex boundary functions.

For a radial conductivity $\gamma(x) = \gamma(|x|)$ in the unit disk $\Omega = \mathbb{D}$, the orthonormal basis functions $\phi_n(\theta) = (2\pi)^{-1/2} e^{in\theta} \in L^2(\partial\Omega)$ are eigenfunctions of the Neumann-to-Dirichlet map, that is

$$(\Lambda_\gamma - \Lambda_1)\phi_n = \lambda_n \phi_n \quad \Rightarrow \quad \langle (\Lambda_\gamma - \Lambda_1)\phi_n, \phi_n \rangle = \lambda_n \quad n = 1, 2, \dots,$$

with λ_n denoting the n 'th eigenvalue of $\Lambda_\gamma - \Lambda_1$. The inverse problem is then to find γ from all eigenvalues $\{\lambda_n\}$.

The corresponding linearized problem is (using (1))

$$\begin{aligned} ((Lh)\phi_n, \phi_n)_{L^2(\partial\Omega)} &= \lambda_n & n = 1, 2, \dots \\ \Leftrightarrow \quad - \int_{\Omega} h |\nabla v_n|^2 dx &= \lambda_n & n = 1, 2, \dots \end{aligned} \quad (2)$$

with

$$\Delta v_n = 0 \text{ in } \Omega, \quad \partial_r v_n = \phi_n \text{ on } \partial\Omega. \quad (3)$$

- (a) Show that that $v_n(r, \theta) = \frac{1}{n} r^n \phi_n(\theta)$ solves (3); show further that

$$|\nabla v_n|^2 = \frac{1}{\pi} r^{2(n-1)} \quad n = 1, 2, \dots$$

Hint: In Cartesian coordinates, $v_n(x, y) = \frac{1}{n}(x + iy)^n$.

- (b) Suppose we have measured the eigenvalues λ_n for $n = 1, 2, \dots$. Then the linearized radial Calderón problem is defined through the operator K as follows

$$K h(n) = -2 \int_0^1 h(r) r^{2n-1} dr = \lambda_n. \quad (4)$$

Show by integrating in polar coordinates that (2) reduce to (4).

- (c) The equation (4) can be discretized (using numerical quadrature) in the form of a matrix $A \in \mathbb{R}^{N \times M}$ with h sampled uniformly in M points along $[0, 1]$ to a vector with N measured eigenvalues. We thereby obtain

$$Ax = y, \quad (5)$$

with $x = [h(r_j)]_{j=1}^M$ and $y = [\lambda_n]_{n=1}^N$.

The discrete problem (5) is implemented in the file `linearEITradial.m` to be found on learn. Download the file, take a look, understand what goes on. In learn you also find the file `b.mat`. They contain eigenvalues of $(\Lambda_\gamma - \Lambda_1)$ in a concentric case.

Solve the linearized inverse problem using your favourite regularization method with sensible choice of regularization parameter (e.g. Morozov), and plot the profile of reconstruction $\hat{\gamma} = h + 1$.

- (d) Assume that the profile is concentric (like last week). Solve the problem as you did last week (by a “direct method”) and compare the solutions.
- (e) How would you generalize the above to the case of a general conductivity that is not radially symmetric?